MURU-WUR-v2: Analytical Chemistry Review Packet

For a qualified external reviewer. Prepared, not sent. The Phase 6 metadata audit is technical evidence collection, not human adjudication.

Study and unresolved point

The frozen MURU-WUR-v2 evaluation selected MS2 scans with numeric collision-energy settings 20 and 60. The project protocol called these NCE. The MSnLib paper reports ID-X MS2 settings of 20 eV and 60 eV. Thermo documentation distinguishes normalized percentage from absolute-voltage settings. The frozen software selects by numeric header value and passes the number through an empirical affine adapter; there is no physical eV-to-NCE conversion.

Source record and checks

- Frozen source identifiers: nine ZIP families declared in src/muru/wur_v2/msnlib_design.py::ZIPS and selected scan membership in artifacts/wur_v2_confirmation_v2/freeze/validation_scan_allowlist.csv.
- Converted source: Zenodo record 15683784, version 5, DOI 10.5281/zenodo.15683784. It describes converted flow-injection mzML from Orbitrap ID-X compound libraries.
- Primary study: Brungs et al., Nature Methods (2025), DOI 10.1038/s41592-025-02813-0; its Data Acquisition methods describe ID-X MS2 energy settings as 20 and 60 eV.
- Instrument documentation: Thermo Orbitrap ID-X Method Editor Help distinguishes normalized collision energy percentage from absolute collision-energy voltage.
- Phase 6 extraction: ../metadata/RAW_METADATA_EXTRACTION.py uses ZIP byte-range reads, verifies member CRC and frozen SHA-256, locates selected spectrum IDs, and extracts mzML instrument/source/software and activation CV terms. The complete all-population validation outcome is in ../metadata/EXTRACTION_VALIDATION_REPORT.md.
- Linked RAW source: Zenodo 13785391 has seven positive-mode RAW families. Its ZIP central directories include 161 .meth files across four source families (67 MCEBIO, 54 NIHNP, 1 MCESCAF, 39 MCEDRUG). Parsed fields identify all 161 as Vanquish_H SeparationMethod files; none exposes an explicit ID-X/NCE/collision-energy setting or a run-level link to selected MS2 spectra. These LC methods do not establish the ID-X MS energy mode. The MassIVE MSV000094528 listing contains 18,281 files and has no method-like extension. Of 1,428 mzML sourceFile RAW names, 835 have a normalized filename candidate, which is not proof of identity. One MCEBIO RAW file (31,994,595 bytes) was streamed and its SHA-1 matched the checksum embedded in the mzML sourceFile record. This establishes identity for that file only. A different filename-associated Zenodo RAW object mismatched; those objects are distinct. The two later families have no candidates in the MassIVE listing. Native methods remain unrecovered.

Frozen processing chain

1. Frozen selector recognizes numeric header values 20 and 60 without decoding their physical units.
2. The measured μ values and model predictions are frozen; Phase 6 does not change spectrum processing or metrics.
3. Candidate and reference are evaluated at the same measured endpoints using the fixed adapter $E=(c-a)/b$, where $a=-5.95552603907965$ and $b=0.8618030610784555$.
4. The affine map is a computational coordinate adapter. No source record establishes it as an eV-to-NCE conversion or an instrument calibration.

Statistical outcome and boundary

1,794 compounds were measured; five failed either model's profile support; 1,789 were scored in 1,687 scaffolds. Candidate P1=0.1296115107, reference P1=0.1335276112, ratio=0.9706719801, paired scaffold-bootstrap 95% interval [0.9523611880, 0.9892672062]. The 2.9328% relative reduction did not pass the predefined ratio≤0.95 practical target. Tail improvement was not demonstrated; the least-similar novelty quartile had ratio approximately 1.0083. The absolute-agreement A0 gate failed; Design A remains INTERFACE_UNRESOLVED.

The comparison is numerically reproducible conditional on the fixed adapter and frozen outputs. It does not establish physical energy-aligned cross-instrument transfer, calibrated collision-energy equivalence, or universal performance. Full raw mzML-to-model reconstruction is not supported by the current package.

Questions for the specialist

Please cite the records/manual version behind each answer.

- Q1. Do the selected converted mzML records justify calling the annotated values 20 and 60 eV, while qualifying original-method provenance?
- Q2. What native Xcalibur method fields distinguish absolute voltage from normalized energy for the relevant software version?
- Q3. What evidence is required to show ProteoWizard copied rather than transformed the energy metadata?
- Q4. Does the one verified MCEBIO RAW-to-mzML SHA-1 match change the evidence standard for other files? What source-specific mapping or native-method records are still needed?
- Q5. Are the paper-level method statement and later v5 library families sufficiently connected for source-specific claims?
- Q6. Does the fixed affine adapter support any physical interpretation, or must it remain only a numeric model coordinate?
- Q7. Is the paired candidate/reference score comparison interpretable under the fixed coordinate, given that the A0 absolute-agreement gate failed?
- Q8. Is the precursor-including mass-weighted endpoint μ chemically meaningful for this bounded question, and what does it omit?
- Q9. Are the energy inputs to full-spectrum comparator models sufficiently fair and transparent for the conditional μ comparison?
- Q10. What exact records or matched instrument experiment would be necessary for a physical transfer claim?

Review status: no human review has occurred. Native method and RAW mapping request drafts are in ../external_requests/; they have not been sent.